

LEMP Stack

+ WordPress

UBUNTU 22.04 · WSL2 · LOCAL DEVELOPMENT

Nginx

MariaDB

PHP 8.1

WordPress

Update the system

Sync Ubuntu's package manager with the latest software sources before installing anything.

```
sudo apt update && sudo apt upgrade -y
```

When prompted for a password, type the one you created at Ubuntu setup. No characters will appear – this is normal.

Install & start Nginx

Nginx handles all incoming HTTP requests and serves your files to the browser.

```
sudo apt install nginx -y
sudo service nginx start
```

VERIFY

Open your Windows browser → **http://localhost**
You should see: *Welcome to nginx!*

Install & secure MariaDB

MariaDB is your database server – a drop-in MySQL replacement that stores all WordPress data.

INSTALL & START

```
sudo apt install mariadb-server -y
sudo service mariadb start
```

SECURE

```
sudo mysql_secure_installation
# Press Enter for root pw
# Set a root password
# Answer Y to all prompts
```

Install PHP-FPM

Nginx doesn't process PHP directly – it passes requests to PHP-FPM, a dedicated process manager.

```
sudo apt install php-fpm php-mysql -y
php -v
# Note the version (e.g. 8.1)
sudo service php8.1-fpm start
# Replace 8.1 with your actual version
```

Link Nginx to PHP

Edit the Nginx config to enable PHP processing, then test and restart the server.

```
sudo nano /etc/nginx/sites-available/default
```

ADD / EDIT THESE BLOCKS

```
index index.php index.html index.htm;

location ~ \.php$ {
    include snippets/fastcgi-php.conf;
    fastcgi_pass unix:/var/run/php/php8.1-fpm.sock;
}
```

```
sudo nginx -t && sudo service nginx restart
```

Verify PHP is working

Create a one-line test file to confirm Nginx and PHP-FPM are communicating correctly.

```
sudo nano /var/www/html/info.php
```

FILE CONTENTS

```
<?php phpinfo(); ?>
```

VERIFY

Open your browser → **http://localhost/info.php**
A large PHP configuration page should load.

502 Bad Gateway? Run `sudo service php8.1-fpm start` and check the socket version in your nginx config matches `php -v`.

Create the WordPress database

Provision an isolated database and a dedicated user account before downloading WordPress.

```
sudo mysql -u root -p
```

RUN INSIDE MARIADB PROMPT

```
CREATE DATABASE wordpress_db;
CREATE USER 'wp_user'@'localhost'
  IDENTIFIED BY 'your_secure_password';
GRANT ALL PRIVILEGES ON wordpress_db.*
  TO 'wp_user'@'localhost';
FLUSH PRIVILEGES;
EXIT;
```

Download & deploy WordPress

Fetch the latest WordPress package and move it into Nginx's public web root.

```
cd /tmp
wget https://wordpress.org/latest.tar.gz
tar -xvf latest.tar.gz
sudo mv wordpress /var/www/html/wordpress
```

SET PERMISSIONS

```
sudo chown -R www-data:www-data /var/www/html/wordpress
sudo chmod -R 755 /var/www/html/wordpress
```

Point Nginx to WordPress

Update the server config so Nginx serves files from the WordPress directory with clean URL support.

```
sudo nano /etc/nginx/sites-available/default
```

UPDATE ROOT PATH

```
root /var/www/html/wordpress;
```

UPDATE LOCATION / BLOCK

```
location / {
    try_files $uri $uri/ /index.php?$args;
}
```

```
sudo nginx -t && sudo service nginx restart
```

Complete setup in browser

Open `http://localhost` in your Windows browser and enter these credentials when prompted.

DATABASE NAME	USERNAME
wordpress_db	wp_user
DATABASE HOST	TABLE PREFIX
localhost	wp_

